

Сетевые технологии

Лабораторная работа №7

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Раздел 1

1. Информация

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Раздел 2

2. Цель работы

2.1 Цель работы

Целью данной лабораторной работы является получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Раздел 3

3. Задание

3.1 Задание

1. Настроить DHCP в случае IPv4

3.1 Задание

1. Настроить DHCP в случае IPv4
2. Настроить DHCP в случае IPv6

Раздел 4

4. Теоретическое введение

4.1 Теоретическое введение

В современных компьютерных сетях для обеспечения корректной работы устройств необходима настройка сетевых параметров, таких как IP-адрес, маска подсети, шлюз по умолчанию и адреса DNS-серверов. Ручная конфигурация этих параметров на каждом узле является трудоёмкой и подверженной ошибкам, поэтому в сетях IPv4 и IPv6 широко применяется автоматическая настройка с использованием протоколов динамической конфигурации.

4.2 Теоретическое введение

Для сетей IPv4 таким механизмом является DHCP (Dynamic Host Configuration Protocol). Он реализует модель клиент–сервер и обеспечивает автоматическое распределение IP-адресов из заранее определённого пула. Передача DHCP-сообщений осуществляется через протокол UDP: сервер принимает запросы клиентов на порту 67 и отправляет ответы на порт 68. Процесс аренды адреса включает обмен сообщениями DHCPDISCOVER, DHCPOFFER, DHCPREQUEST и DHCPACK.

4.3 Теоретическое введение

В сетях IPv6 применяются два механизма автоматической настройки: SLAAC (Stateless Address Autoconfiguration) и DHCPv6. SLAAC позволяет узлу самостоятельно сформировать IPv6-адрес на основе информации, поступающей от маршрутизатора в виде ICMPv6-сообщений Router Advertisement. Проверка уникальности созданного адреса выполняется с помощью процедуры обнаружения дубликатов (DAD). Однако SLAAC предоставляет только базовые параметры, поэтому для получения дополнительной информации может использоваться DHCPv6.

Раздел 5

5. Выполнение лабораторной работы

5.1 Настройка DHCP в случае IPv4

Запустим GNS3 VM и GNS3. Создадим новый проект. В рабочем пространстве разместим и соединим устройства в соответствии с топологией данной в лабораторной работе. Изменим название устройств, указав наш логин и счетный номер.

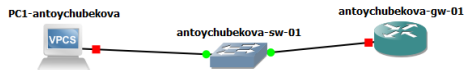


Рисунок 1: Топология моделируемой сети

5.2 Настройка DHCP в случае IPv4

Включим захват трафика на соединении между коммутатором sw-01 и маршрутизатором gw-01.

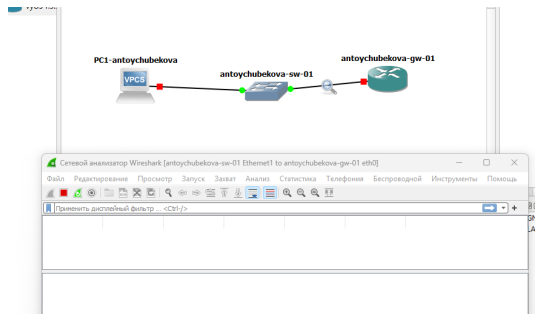


Рисунок 2: Захват трафика

5.3 Настройка DHCP в случае IPv4

На маршрутизаторе перейдем в режим конфигурирования, изменим имя устройства и доменное имя, заменим системного пользователя, заданного по умолчанию, на нашего пользователя.

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name antoychubekova-gw-01
[edit]
vyos@vyos# set system domain-name antoychubekova.net
[edit]
vyos@vyos# set system login user antoychubekova authentication plaintext-password 123456
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout

Welcome to VyOS - antoychubekova-gw-01 tty50
antoychubekova-gw-01 login: █
```

Рисунок 3: Изменение системного пользователя

5.4 Настройка DHCP в случае IPv4

Зайдем под нашим пользователем и удалим пользователя vyos.

```
antoychubekova-gw-01 login: antoychubekova
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
antoychubekova@antoychubekova-gw-01:~$ configure
[edit]
antoychubekova@antoychubekova-gw-01# delete system login user vyos
[edit]
antoychubekova@antoychubekova-gw-01# commit
[edit]
antoychubekova@antoychubekova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
antoychubekova@antoychubekova-gw-01#
```

Рисунок 4: Удаление пользователя vyos

5.5 Настройка DHCP в случае IPv4

На маршрутизаторе под созданным пользователем перейдем в режим конфигурирования и настроим адресацию IPv4.

```
antoychubekova@antoychubekova-gw-01# set interfaces ethernet eth0 address 10.0.0
.1/24
[edit]
antoychubekova@antoychubekova-gw-01#
```

Рисунок 5: Настройка адресации IPv4

5.6 Настройка DHCP в случае IPv4

Добавим конфигурацию DHCP-сервера на маршрутизаторе. Создадим разделяемую сеть (shared-network-name) с названием antoychubekova.

```
antoychubekova@antoychubekova-gw-01# set service dhcp-server shared-network-name
antoychubekova domain-name antoychubekova.net
[edit]
antoychubekova@antoychubekova-gw-01# set service dhcp-server shared-network-name
antoychubekova name-server 10.0.0.1
[edit]
```

Рисунок 6: Добавление конфигурации DHCP-сервера

5.7 Настройка DHCP в случае IPv4

Далее создадим подсеть (subnet) с адресом 10.0.0.0/24, задан диапазон адресов (range) с именем hosts, содержащий адреса 10.0.0.2–10.0.0.253.

```
antoychubekova@antoychubekova-gw-01# set service dhcp-server shared-network-name
  antoychubekova subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
antoychubekova@antoychubekova-gw-01# set service dhcp-server shared-network-name
  antoychubekova subnet 10.0.0.0/24 range hosts start 10.0.0.2
[edit]
antoychubekova@antoychubekova-gw-01# set service dhcp-server shared-network-name
  antoychubekova subnet 10.0.0.0/24 range hosts stop 10.0.0.253
[edit]
```

Рисунок 7: Добавление конфигурации DHCP-сервера

5.8 Настройка DHCP в случае IPv4

Сохраним наши изменения.

```
[edit]  
antoychubekova@antoychubekova-gw-01# commit  
[edit]  
antoychubekova@antoychubekova-gw-01# save  
Saving configuration to '/config/config.boot'...  
Done  
[edit]  
antoychubekova@antoychubekova-gw-01# exit  
exit  
antoychubekova@antoychubekova-gw-01:~$
```

Рисунок 8: Сохранения изменений

5.9 Настройка DHCP в случае IPv4

Просмотрим статистики DHCP-сервера и выданных адресов. Мы видим, что сводную статистику DHCP-сервера: размер пула адресов, количество выданных аренд (leases), доступные адреса и процент использования пула. В представленном выводе пул содержит 252 адреса, все они доступны, аренды отсутствуют (0% использования). Также видим выданные IP-адреса, MAC-адреса клиентов, состояние аренды, время начала и окончания, оставшееся время, пул и имя хоста. В данном выводе аренды отсутствуют, поэтому таблица пуста.

5.10 Настройка DHCP в случае IPv4

```
antoychubekova@antoychubekova-gw-01:~$ show dhcp server statistics
Pool           Size    Leases    Available  Usage
-----
antoychubekova    252      0        252      0%
antoychubekova@antoychubekova-gw-01:~$ show dhcp server leases
IP address      Hardware address  State    Lease start    Lease expiration  Re
maining        Pool      Hostname
-----
antoychubekova@antoychubekova-gw-01:~$
```

Рисунок 9: Статистика DHCP-сервера

5.11 Настройка DHCP в случае IPv4

Настроим оконечное устройство PC1.

```
VPCS> ip dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = antoychubekova.net
```

Рисунок 10: Настройка оконечного устройства

5.12 Настройка DHCP в случае IPv4

```
Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00
Option 12: Host Name = VPCS

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = antoychubekova.net

IP 10.0.0.2/24 GW 10.0.0.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> █
```

Рисунок 11: Настройка оконечного устройства

5.13 Настройка DHCP в случае IPv4

Проверим конфигурацию IPv4 на узле, пропингуем маршрутизатор. Мы видим, что все корректно работает.

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.0.0.2/24
GATEWAY     : 10.0.0.1
DNS         : 10.0.0.1
DHCP SERVER : 10.0.0.1
DHCP LEASE  : 86220, 86400/43200/75600
DOMAIN NAME : antoychubekova.net
MAC         : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

VPCS> ping 10.0.0.1 -c 2

84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=2.686 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=5.191 ms

VPCS> 
```

Рисунок 12: Проверка конфигурации IPv4

5.14 Настройка DHCP в случае IPv4

На маршрутизаторе вновь посмотрим статистику DHCP-сервера и выданные адреса.

```
antoychubekova@antoychubekova-gw-01:~$ show dhcp server statistics
Pool          Size      Leases    Available  Usage
-----
antoychubekova 252        1         251      0%
antoychubekova@antoychubekova-gw-01:~$ show dhcp server leases
IP address    Hardware address  State    Lease start    Lease expiration  Remaining
Pool          Hostname
-----
10.0.0.2      00:50:79:66:68:00 active    2025/12/03 07:30:50 2025/12/04 07:30:50 23:55:39
antoychubekova VPCS
antoychubekova@antoychubekova-gw-01:~$
```

Рисунок 13: Статистика DHCP-сервера

5.15 Настройка DHCP в случае IPv4

На маршрутизаторе посмотрим журнал работы DHCP-сервера.

```

antonychubekova@antonychubekova-pc-01:~$ show log | grep dhcp
Dec 03 06:53:08 dhclient-script-vyos[1818]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:53:15 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 06:53:16 dhclient-script-vyos[1818]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:53:22 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 06:54:42 dhclient-script-vyos[2192]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:54:48 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 06:54:50 dhclient-script-vyos[2192]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:55:01 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 06:56:43 dhclient-script-vyos[2501]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:56:44 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 06:56:46 dhclient-script-vyos[2501]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:56:45 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 06:58:55 dhclient-script-vyos[2693]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:58:56 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 06:58:56 dhclient-script-vyos[2693]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 06:59:27 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:01:27 dhclient-script-vyos[2741]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:01:28 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:01:28 dhclient-script-vyos[2741]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:01:28 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:03:46 dhclient-script-vyos[2839]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:03:47 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:03:47 dhclient-script-vyos[2839]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:03:48 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:06:18 dhclient-script-vyos[3315]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:06:19 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:06:20 dhclient-script-vyos[3315]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:06:21 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:08:47 dhclient-script-vyos[3376]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:08:49 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:08:50 dhclient-script-vyos[3376]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:08:51 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:10:36 dhclient-script-vyos[3426]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:10:39 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:10:39 dhclient-script-vyos[3426]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:10:39 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:12:58 dhclient-script-vyos[3500]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:13:10 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:13:11 dhclient-script-vyos[3500]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:13:12 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:15:26 dhclient-script-vyos[3556]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:15:28 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:15:28 dhclient-script-vyos[3556]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:15:29 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Dec 03 07:16:54 sudo[3698]: antonychubekova : TTY=ttty0 : FWD=/home/antonychubekova : USER=root : COMMAND=/usr/bin/sh -c
/usr/sbin/vysshim /usr/libexec/vyos/conf_mode/dhcp_server.py
Dec 03 07:16:54 vyos-configd[679]: Received message: {"type": "mode", "data": {"usr/libexec/vyos/conf_mode/dhcp_server.py}}

```

Рисунок 14: Журнал работы DHCP-сервера

5.16 Настройка DHCP в случае IPv4

```

Dec 03 07:17:09 dhclient-script-vyos[3732]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:17:11 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:17:12 dhclient-script-vyos[3732]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:17:23 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:19:20 dhclient-script-vyos[3773]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:19:23 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:19:23 dhclient-script-vyos[3773]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:19:23 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:21:04 dhclient-script-vyos[3809]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:21:06 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:21:06 dhclient-script-vyos[3809]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:21:07 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:23:39 dhclient-script-vyos[3872]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:23:49 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:23:49 dhclient-script-vyos[3872]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:23:57 vyos-hosted[680]: Request data: {"type": "name_servers", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:24:15 dhclient-script-vyos[3940]: Deleting search domains with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:24:17 vyos-hosted[680]: Request data: {"type": "search_domains", "op": "delete", "data": ["dhcp-eth0"]}
Dec 03 07:24:18 dhclient-script-vyos[3940]: Deleting nameservers with tag "dhcp-eth0" via vyos-hosted-client
Dec 03 07:24:32 sudo[4048]: antoychubekova : TTY=ttty50 ; FWD=/home/antoychubekova : USER=root : COMMAND=/usr/bin/sh -c
/usr/sbin/vysshm /usr/libexec/vyos/conf_mode/dhcp_server.py
Dec 03 07:24:41 dhcpd[4066]: Wrote 0 leases to leases file.
Dec 03 07:24:41 dhcpd[4066]: Lease file test successful, removing temp lease file: /config/dhcpd.leases.1764746681
Dec 03 07:24:41 dhcpd[4066]: Wrote 0 leases to leases file.
Dec 03 07:24:41 dhcpd[4066]:
Dec 03 07:24:41 dhcpd[4066]: No subnet declaration for eth2 (no IPv4 addresses).
Dec 03 07:24:41 dhcpd[4066]: ** Ignoring requests on eth2. If this is not what
Dec 03 07:24:41 dhcpd[4066]: you want, please write a subnet declaration
Dec 03 07:24:41 dhcpd[4066]: in your dhcpd.conf file for the network segment
Dec 03 07:24:41 dhcpd[4066]: to which interface eth2 is attached. **
Dec 03 07:24:41 dhcpd[4066]:
Dec 03 07:24:41 dhcpd[4066]: No subnet declaration for eth1 (no IPv4 addresses).
Dec 03 07:24:41 dhcpd[4066]: ** Ignoring requests on eth1. If this is not what
Dec 03 07:24:41 dhcpd[4066]: you want, please write a subnet declaration
Dec 03 07:24:41 dhcpd[4066]: in your dhcpd.conf file for the network segment
Dec 03 07:24:41 dhcpd[4066]: to which interface eth1 is attached. **
Dec 03 07:24:41 dhcpd[4066]:
Dec 03 07:24:42 dhcpd[4066]: Server starting service.
Dec 03 07:27:25 sudo[4141]: antoychubekova : TTY=ttty50 ; FWD=/home/antoychubekova : USER=root : COMMAND=/usr/libexec/vy
os/op_mode/show_dhcp.py --statistics
Dec 03 07:28:22 sudo[4167]: antoychubekova : TTY=ttty50 ; FWD=/home/antoychubekova : USER=root : COMMAND=/usr/libexec/vy
os/op_mode/show_dhcp.py --leases
Dec 03 07:30:46 dhcpd[4066]: DHCPDISCOVER from 00:50:79:66:68:00 via eth0
Dec 03 07:30:49 dhcpd[4066]: DHCPOFFER on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0
Dec 03 07:30:50 dhcpd[4066]: DHCPREQUEST for 10.0.0.2 (10.0.0.1) from 00:50:79:66:68:00 (VPCS) via eth0
Dec 03 07:30:50 dhcpd[4066]: DHCPACK on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0
Dec 03 07:34:55 sudo[4194]: antoychubekova : TTY=ttty50 ; FWD=/home/antoychubekova : USER=root : COMMAND=/usr/libexec/vy
os/op_mode/show_dhcp.py --statistics
Dec 03 07:35:07 sudo[4200]: antoychubekova : TTY=ttty50 ; FWD=/home/antoychubekova : USER=root : COMMAND=/usr/libexec/vy
os/op_mode/show_dhcp.py --leases
antoychubekova@antoychubekova-gw-01:~$

```

Рисунок 15: Журнал работы DHCP-сервера

5.17 Настройка DHCP в случае IPv4

Логи показывают работу DHCP-сервера и связанные с ним процессы в системе VyOS. Видно удаление старых DNS-настроек интерфейса eth0, запуск DHCP-демона, проверку конфигурации, а также ключевые события аренды: получение клиентом DHCPDISCOVER, отправку DHCPOFFER, DHCPREQUEST и подтверждение DHCPACK.

Рисунок 16: DHCP Discover

[illegible]

5.19 Настройка DHCP в случае IPv4

The screenshot displays a network traffic capture in Wireshark. The main window shows a list of captured packets, with the selected packet being a DHCP Discover. The packet details pane on the right provides a hierarchical view of the packet's structure, including the Ethernet II header, the Internet Protocol (IP) header, and the DHCP Discover message. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII format.

No.	Time	Source	Destination	Protocol	Length	Info
72	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
73	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
74	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
75	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
76	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
77	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
78	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
79	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
80	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
81	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
82	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
83	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
84	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
85	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
86	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
87	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
88	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
89	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
90	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
91	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
92	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
93	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
94	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
95	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
96	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
97	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
98	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
99	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
100	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
101	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
102	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
103	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
104	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
105	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
106	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
107	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
108	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
109	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
110	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
111	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
112	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
113	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
114	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
115	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
116	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction
117	0.000000	192.168.1.100	255.255.255.255	DHCP	342	DHCP Discover - Transaction

Frame 101: 406 bytes on wire (3248 bits), 406 bytes captured (3248 bits) on interface 0, 14 B captured on interface 0 (eth0)

Ethernet II, Src: Intel(R) Ethernet Controller (08:00:27:00:00:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Internet Protocol Version 4, Src: 192.168.1.100, Dst: 255.255.255.255

DHCP Discover

... 0000 ... > Header Length: 20 bytes (1)

Total Length: 302

Differentiated Services Field: DS (DSCP: Unknown, ECN: Not-ECT)

Identification: 0x0000 (0)

Flags: none

Fragment Offset: 0

Time to Live: 64

Protocol: UDP (17)

Header Checksum: 0x0000 (validation disabled)

Header checksum status: Invalid

Source Address: 192.168.1.100

Destination Address: 255.255.255.255

User Datagram Protocol, Src Port: 67, Dst Port: 67

Source Port: 67

Destination Port: 67

Length: 302

Checksum: 0x0000 (unverified)

[Checksum Status: Unverified]

[Stream Index: 0]

[Timestamps]

UDP payload (304 bytes)

Dynamic Host Configuration Protocol (Discover)

Message Type: Boot Request (1)

Hardware Type: Ethernet (0x01)

Hardware address length: 6

Hop: 0

Transaction ID: 0x00000000

Seconds elapsed: 0

Bootp flags: 0x0000 (unicast)

Client IP address: 0.0.0.0

Your (client) IP address: 0.0.0.0

Рисунок 17: DHCP Discover

5.20 Настройка DHCP в случае IPv4

Time	Source	Destination	Protocol	Length	Info
72.1783.829861	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
73.1785.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
74.1785.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
75.1785.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
76.1785.178483	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
77.1785.879836	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
78.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
79.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
80.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
81.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
82.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
83.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
84.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
85.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
86.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
87.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
88.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
89.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
90.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
91.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
92.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
93.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
94.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
95.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
96.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
97.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
98.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
99.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
100.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
101.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
102.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
103.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
104.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
105.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
106.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
107.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
108.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
109.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
110.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
111.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
112.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
113.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
114.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
115.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
116.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted
117.1882.180881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover -> Transmitted

Frame 382: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface ..., 5d 8
Interface 0: 8 (1)
Encapsulation type: Ethernet (1)
Arrival Time: Dec 5, 2023 10:58:46.303079000 RTT: 2 (ms)
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 17852861.303079000 seconds
[Time delta from previous captured frame: 0.005700000 seconds]
[Time delta from previous displayed frame: 0.005700000 seconds]
[Time since reference or first frame: 0.054300000 seconds]
Frame Number: 100
Frame Length: 342 bytes (2736 bits)
Capture Length: 342 bytes (2736 bits)
Frame Is marked: false
Frame Is ignored: false
[Protocol is Frame: ethertype:ip:udp:dhcp]
[Following Rule Name: dm]
[Following Rule String: udp]
Ethernet II, Src: RealtekUc180-80, Dest: Private_00:50:56:00:00:00 (00:50:56:00:00:00)
Destination: Private_00:50:56:00:00:00 (00:50:56:00:00:00)
Source: RealtekUc180-80 (00:50:56:00:00:00)
Type: IP (0x0800)
Internet Protocol Version 4, Src: 18.8.8.1, Dest: 18.8.8.2
6508
... > Version: 4
... > Header Length: 20 bytes (5)
Differentiated Services Field: 0x0 (DSCP: Unknown, ECN: Not-ECT)
Total Length: 108
Identification: 0x0000 (0)
Flags: none
Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 18.8.8.1
Destination Address: 18.8.8.2
User Datagram Protocol, Src Port: 67, Dest Port: 68
Source Port: 67
Destination Port: 68
Length: 108
Checksum: 0x0000 [unverified]
[Checksum Status: Unverified]
[Stream Index: 1]
[Timestamp]
UDP payload (100 bytes)
Dynamic Host Configuration Protocol [offer]
Message type: Boot Reply (2)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Magi: 0
Transaction ID: 0x00000000
Seconds elapsed: 0
Resume flag: 0x0000 (Unknown)
Client IP address: 0.0.0.0
Your (client) IP address: 18.8.8.2

Рисунок 18: DHCP Offer

Рисунок 19: DHCP Request

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡

Рисунок 20: DHCP АСК



5.23 Настройка DHCP в случае IPv4

94	2147.356219	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover - Transaction ID 0x6172c85e
95	2157.279441	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover - Transaction ID 0x6172c85e
96	2166.045696	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover - Transaction ID 0x6172c85e
97	2175.580079	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover - Transaction ID 0x6172c85e
98	2185.726876	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover - Transaction ID 0x6172c85e
99	2634.355356	0.0.0.0	255.255.255.255	DHCP	406 DHCP Discover - Transaction ID 0x76d5f79
100	2634.397245	0c:e1:6e:cf:00:00	Broadcast	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
101	2635.350037	0.0.0.0	255.255.255.255	DHCP	406 DHCP Discover - Transaction ID 0x76d5f79
102	2635.415005	10.0.0.1	10.0.0.2	DHCP	342 DHCP Offer - Transaction ID 0x76d5f79
103	2635.420120	0c:e1:6e:cf:00:00	Broadcast	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
104	2636.443829	0c:e1:6e:cf:00:00	Broadcast	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
105	2638.356983	0.0.0.0	255.255.255.255	DHCP	406 DHCP Request - Transaction ID 0x76d5f79
106	2638.422637	10.0.0.1	10.0.0.2	DHCP	342 DHCP ACK - Transaction ID 0x76d5f79
107	2639.358929	Private_66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (Request)
108	2640.359045	Private_66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (Request)
109	2641.361506	Private_66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.2 (Request)
110	2849.239663	Private_66:68:00	Broadcast	ARP	64 Who has 10.0.0.1? Tell 10.0.0.2
111	2849.242725	0c:e1:6e:cf:00:00	Private_66:68:00	ARP	60 10.0.0.1 is at 0c:e1:6e:cf:00:00
112	2849.242785	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) request id=0xfce7, seq=1/256, ttl=64 (reply in 113)
113	2849.244751	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) reply id=0xfce7, seq=1/256, ttl=64 (request in 112)
114	2850.247477	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) request id=0xfde7, seq=2/512, ttl=64 (reply in 115)
115	2850.251144	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) reply id=0xfde7, seq=2/512, ttl=64 (request in 114)
116	2054.325987	0c:e1:6e:cf:00:00	Private_66:68:00	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
117	2054.328143	Private_66:68:00	0c:e1:6e:cf:00:00	ARP	60 10.0.0.2 is at 00:50:79:66:68:00

Рисунок 21: Полный процесс динамического получения IP-адреса клиентом через DHCP

5.24 Настройка DHCP в случае IPv6

В предыдущем проекте в рабочем пространстве дополним сеть, разместив и соединив устройства в соответствии с топологией, приведённой в лабораторной работе. Используем хост (клиент) Kali Linux CLI, поскольку клиент VPCS не поддерживает DHCPv6. Измените отображаемые названия устройств.

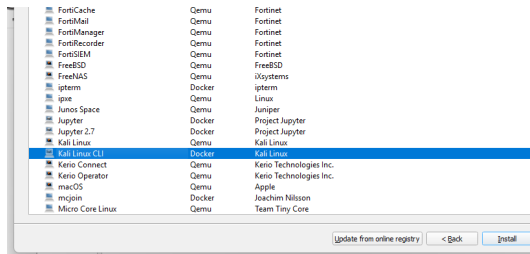


Рисунок 22: Добавление Kali Linux CLI

5.25 Настройка DHCP в случае IPv6

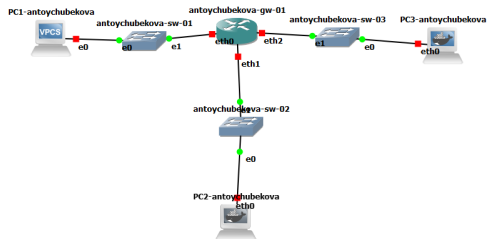


Рисунок 23: Топология моделируемой сети

5.26 Настройка DHCP в случае IPv6

Включим захват трафика на соединениях между маршрутизатором gw-01 и коммутаторами sw-02 и sw-03.

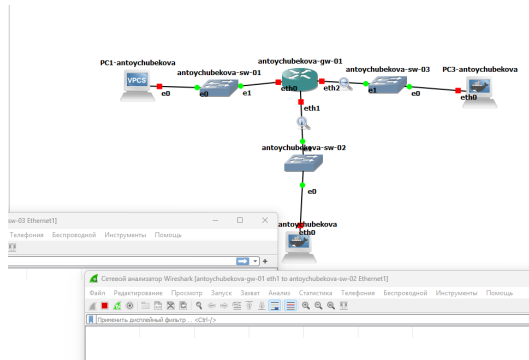


Рисунок 24: Включение захвата трафика

5.27 Настройка DHCP в случае IPv6

Настроим адресацию IPv6 на маршрутизаторе.

```
antoychubekova-gw-01 login: antoychubekova
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
antoychubekova@antoychubekova-gw-01:~$ configure
[edit]
antoychubekova@antoychubekova-gw-01# set interfaces ethernet eth1 address 2000::
1/64
[edit]
antoychubekova@antoychubekova-gw-01# set interfaces ethernet eth2 address 2001::
1/64
[edit]
antoychubekova@antoychubekova-gw-01# show interfaces
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:e1:6e:cf:00:00
  }
  ethernet eth1 {
+   address 2000::1/64
    hw-id 0c:e1:6e:cf:00:01
  }
  ethernet eth2 {
+   address 2001::1/64
    hw-id 0c:e1:6e:cf:00:02
  }
  loopback lo {
  }
[edit]
antoychubekova@antoychubekova-gw-01#
```

5.28 Настройка DHCP в случае IPv6

Сохраним наши изменения.

```
antoychubekova@antoychubekova-gw-01# commit  
[edit]  
antoychubekova@antoychubekova-gw-01# save  
Saving configuration to '/config/config.boot'...  
Done  
[edit]  
antoychubekova@antoychubekova-gw-01#
```

Рисунок 26: Сохранение изменений

5.29 Настройка DHCP в случае IPv6

На маршрутизаторе настроим DHCPv6 без отслеживания состояния. Настроим объявления о маршрутизаторах (Router Advertisements, RA) на интерфейсе eth1.

```
antoychubekova@antoychubekova-gw-01# set service router-advert interface eth1 pr  
efix 2000::/64  
[edit]  
antoychubekova@antoychubekova-gw-01# set service router-advert interface eth1 ot  
her-config-flag  
[edit]  
antoychubekova@antoychubekova-gw-01#
```

Рисунок 27: Настройка объявления о маршрутизаторах

5.30 Настройка DHCP в случае IPv6

Добавим конфигурации DHCP-сервера. Создаем разделяемую сеть (shared-network-name) с названием antoychubekova, задаем информацию общих опций (common-options) для разделяемой сети. При этом подсеть (subnet) 2000::/64 не требуется настраивать, поскольку она не будет содержать полезной информации.

```
antoychubekova@antoychubekova-gw-01# set service dhcpv6-server shared-network-name antoychubekova-stateless
[edit]
antoychubekova@antoychubekova-gw-01# set service dhcpv6-server shared-network-name antoychubekova-stateless subnet 2000::0/64
[edit]
antoychubekova@antoychubekova-gw-01# set service dhcpv6-server shared-network-name antoychubekova-stateless common-options name-server 2000::1
[edit]
antoychubekova@antoychubekova-gw-01# set service dhcpv6-server shared-network-name antoychubekova-stateless common-options domain-search antoychubekova.net
[edit]
antoychubekova@antoychubekova-gw-01#
```

Рисунок 28: Добавление конфигурации DHCP-сервера

5.31 Настройка DHCP в случае IPv6

Далее сохраняем наши изменения и посмотрим корректность отработки выше команд.

```
antoychubekova@antoychubekova-gw-01# commit
[edit]
antoychubekova@antoychubekova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
antoychubekova@antoychubekova-gw-01# run show configuration
interfaces {
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:el:6e:cf:00:00
  }
  ethernet eth1 {
    address 2000::1/64
    hw-id 0c:el:6e:cf:00:01
  }
  ethernet eth2 {
    address 2001::1/64
    hw-id 0c:el:6e:cf:00:02
  }
  loopback lo {
  }
}
service {
  dhcp-server {
    shared-network-name antoychubekova {
      domain-name antoychubekova.net
      name-server 10.0.0.1
      subnet 10.0.0.0/24 {
        default-router 10.0.0.1
        range hosts {
          start 10.0.0.2
          stop 10.0.0.253
        }
      }
    }
  }
  dhcpv6-server {
    shared-network-name antoychubekova-stateless {
      common-options {
        domain-search antoychubekova.net
        name-server 2000::1
      }
      subnet 2000::0/64 {
      }
    }
  }
}
```

5.32 Настройка DHCP в случае IPv6

```
system {
  config-management {
    commit-revisions 100
  }
  conntrack {
    modules {
      ftp
      h323
      nfs
      pptp
      sip
      sqlnet
      tftp
    }
  }
  console {
    device ttyS0 {
      speed 115200
    }
  }
  domain-name antoychubekova.net
  host-name antoychubekova-gw-01
  login {
    user antoychubekova {
      authentication {
        encrypted-password *****
      }
    }
  }
  ntp {
    server time1.vyos.net {
    }
    server time2.vyos.net {
    }
    server time3.vyos.net {
    }
  }
  syslog {
    global {
      facility all {
        level info
      }
      facility protocols {
        level debug
      }
    }
  }
}
```

5.33 Настройка DHCP в случае IPv6

На узле PC2 проверим настройки сети. На ПК2 настроены два сетевых интерфейса: eth0 с глобальным IPv6-адресом 2000::3c40:8bff:fece:ad0a и локальным fe80::3c40:8bff:fece:ad0a, а также eth1 с локальным IPv6-адресом fe80::b0ee:9cff:fea5:a8. Интерфейс lo настроен для loopback (127.0.0.1 и ::1).

Маршрутизация IPv6 реализована через eth0 для глобальной сети 2000::/64 и через локальные адреса для link-local сети fe80::/64. Шлюз по умолчанию (::/0) настроен через адрес fe80::0c:e1:6e:ff:fe:cf:01 на eth0.

5.34 Настройка DHCP в случае IPv6

```
(root@PC2-antoychubekova)~#  
# ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::3c40:8bff:fece:ad0a prefixlen 64 scopeid 0x20<link>  
    inet6 2000::3c40:8bff:fece:ad0a prefixlen 64 scopeid 0x0<global>  
    ether 3e:40:8b:ce:ad:0a txqueuelen 1000 (Ethernet)  
    RX packets 3 bytes 282 (282.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 9 bytes 766 (766.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::b0ee:9cff:fea5:a8 prefixlen 64 scopeid 0x20<link>  
    ether b2:ee:9c:a5:00:a8 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(root@PC2-antoychubekova)~#
```

Рисунок 31: Настройка сети

5.35 Настройка DHCP в случае IPv6

```
(root@PC2-antoychubekova)~]
# route -n -A inet6
Kernel IPv6 routing table
Destination                Next Hop                    Flag Met Ref  Use If
2000::/64                   ::                          UAe  256 1    0 eth0
fe80::/64                   ::                          U    256 1    0 eth0
fe80::/64                   ::                          U    256 1    0 eth1
::/0                        fe80::eel:6eff:feof:1      UGDae 1024 1    0 et
h0
::1/128                     ::                          Un   0  3    0 lo
2000::3c40:8bff:fece:ad0a/128 ::                          Un   0  2    0 eth0
fe80::3c40:8bff:fece:ad0a/128 ::                          Un   0  5    0 eth0
fe80::b0ee:9cff:fea5:a8/128  ::                          Un   0  3    0 eth1
ff00::/8                    ::                          U    256 2    0 eth0
ff00::/8                    ::                          U    256 1    0 eth1
::/0                        ::                          !n   -1 1    0 lo

(root@PC2-antoychubekova)~]
#
```

Рисунок 32: Настройка сети

5.36 Настройка DHCP в случае IPv6

На узле PC2 пропингуем маршрутизатор. Мы видим, что все правильно обрабатывается.

```
(root@ PC2-antoychubekova) -[~]
# ping 2000::1 -c 2
PING 2000::1 (2000::1) 56 data bytes
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=14.1 ms
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=3.64 ms

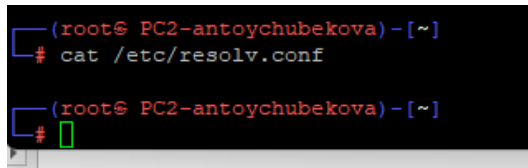
--- 2000::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 3.636/8.860/14.085/5.224 ms

(root@ PC2-antoychubekova) -[~]
#
```

{{width=

5.37 Настройка DHCP в случае IPv6

На узле PC2 проверим настройки DNS.

A terminal window with a black background and red text. The prompt is (root@ PC2-antoychubekova) - [~]. The first command is # cat /etc/resolv.conf. The second command is # followed by a green cursor box.

```
(root@ PC2-antoychubekova) - [~]  
# cat /etc/resolv.conf  
  
(root@ PC2-antoychubekova) - [~]  
#
```

Рисунок 33: Настройка DNS

5.38 Настройка DHCP в случае IPv6

На узле PC2 получим адрес по DHCPv6. Здесь опция -6 указывает на использование протокола DHCPv6, опция -S — на запрос только информации DHCPv6, но не адреса, опция -v — на вывод на экран подробной информации.

```
(root@ PC2-antoychubekova)~[~]
# dhclient -6 -S -v eth0
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on Socket/eth0
Sending on   Socket/eth0
Created duid "\000\003\000\001>@\213\316\255\012".
PRC: Requesting information (INIT).
XMT: Forming Info-Request, 0 ms elapsed.
XMT: Info-Request on eth0, interval 930ms.
RCV: Reply message on eth0 from fe80::eel:6eff:feef:1.
PRC: Done.

(root@ PC2-antoychubekova)~[~]
#
```

Рисунок 34: Настройка DNS

5.39 Настройка DHCP в случае IPv6

Вновь пропингуем от узла PC2 маршрутизатор, проверим настройки DNS. Мы видим имя домена и его IPv6.

```
(root@ PC2-antoychubekova)~  
# ping 2000::1 -c2  
PING 2000::1 (2000::1) 56 data bytes  
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=5.39 ms  
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=4.12 ms  
  
--- 2000::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 100lms  
rtt min/avg/max/mdev = 4.117/4.755/5.393/0.638 ms  
  
(root@ PC2-antoychubekova)~  
# cat /etc/resolv.conf  
search antoychubekova.net.  
nameserver 2000::1  
  
(root@ PC2-antoychubekova)~  
#
```

Рисунок 35: Настройки DNS и проверка доступности

5.40 Настройка DHCP в случае IPv6

На маршрутизаторе посмотрим статистику DHCP-сервера и выданные адреса.

```
antoychubekova@antoychubekova-gw-01# run show dhcpv6 server leases
IPv6 address  State    Last communication    Lease expiration    Remaining
Type    Pool    IAID_DUID
-----
[edit]
```

Рисунок 36: Статистика DHCP

Далее посмотрим захваченный трафик.

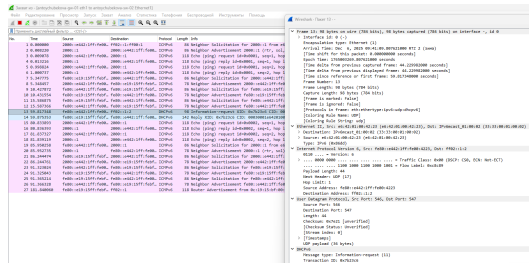


Рисунок 37: DHCP-пакет

5.42 Настройка DHCP в случае IPv6

The screenshot displays a Wireshark packet capture of a DHCPv6 Solicitation. The packet list on the left shows a DHCPv6 Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details on the right show the DHCPv6 Solicitation structure, including the Transaction ID, Message Type (1), and various flags and options.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2001:::442:1ff:f6b6	ff02::1:3	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
2	0.000020	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
3	0.000070	2001:::442:1ff:f6b6	2001:::1	DHCPv6	110	Echo (ping) request 10000000, seqno, hop limit
4	0.000120	2001:::1	2001:::442:1ff:f6b6	DHCPv6	110	Echo (ping) reply 10000000, seqno, hop limit
5	0.000170	2001:::442:1ff:f6b6	2001:::1	DHCPv6	110	Echo (ping) request 10000000, seqno, hop limit
6	0.000220	2001:::1	2001:::442:1ff:f6b6	DHCPv6	110	Echo (ping) reply 10000000, seqno, hop limit
7	0.000270	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
8	0.000320	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
9	0.000370	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
10	0.000420	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
11	0.000470	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
12	0.000520	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
13	0.000570	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
14	0.000620	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
15	0.000670	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
16	0.000720	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
17	0.000770	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
18	0.000820	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
19	0.000870	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
20	0.000920	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
21	0.000970	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
22	0.001020	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
23	0.001070	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
24	0.001120	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
25	0.001170	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1
26	0.001220	2001:::1	2001:::442:1ff:f6b6	DHCPv6	80	Neighbor Advertisement 2001:::1 (IPv6, v6ll, v6v)
27	0.001270	2001:::442:1ff:f6b6	2001:::1	DHCPv6	80	Neighbor Solicitation For 2001:::442:1ff:f6b6:1

The packet details on the right show the DHCPv6 Solicitation structure, including the Transaction ID, Message Type (1), and various flags and options.

Packet 1: DHCPv6 Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 2: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 3: DHCPv6 Echo (ping) request (Type 3) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (3), and various flags and options.

Packet 4: DHCPv6 Echo (ping) reply (Type 4) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (4), and various flags and options.

Packet 5: DHCPv6 Echo (ping) request (Type 3) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (3), and various flags and options.

Packet 6: DHCPv6 Echo (ping) reply (Type 4) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (4), and various flags and options.

Packet 7: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 8: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 9: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 10: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 11: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 12: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 13: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 14: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 15: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 16: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 17: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 18: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 19: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 20: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 21: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 22: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 23: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 24: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 25: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Packet 26: DHCPv6 Neighbor Advertisement (Type 2) from ff02::1:3 to fe80::... The packet details show the Transaction ID, Message Type (2), and various flags and options.

Packet 27: DHCPv6 Neighbor Solicitation (Type 1) from fe80::... to ff02::1:3. The packet details show the Transaction ID, Message Type (1), and various flags and options.

Рисунок 38: DHCPv6-пакет

5.43 Настройка DHCP в случае IPv6

Захват из - [antychubekova-gw-01 eth1 to antychubekova-sw-02 Ethernet]

Файл Редактирование Просмотр Запуск Захват Анализ Статистика Телефония Базовые инструменты Инструменты Помощь

Применить десятичный фильтр: <Ctrl>F

No.	Time	Source	Destination	Protocol	Length	Info
1	0.800800	2000::e442:1fff:fe00...	ff02::1::ff00::1	ICMPv6	86	Neighbor Solicitation for 2000::1 from e6:42:01:00:42:23
2	0.800820	2000::1	2000::e442:1fff:fe00...	ICMPv6	86	Neighbor Advertisement 2000::1 (rtr, sol, ov) Is at 0c:19:15:bf:00:01
3	0.800978	2000::e442:1fff:fe00...	2000::1	ICMPv6	118	Echo (ping) request id=0x0001, seq=1, hop limit=64 (reply in 4)
4	0.813216	2000::1	2000::e442:1fff:fe00...	ICMPv6	118	Echo (ping) reply id=0x0001, seq=1, hop limit=64 (request in 5)
5	0.908824	2000::e442:1fff:fe00...	2000::1	ICMPv6	118	Echo (ping) request id=0x0001, seq=2, hop limit=64 (reply in 6)
6	1.000737	2000::1	2000::e442:1fff:fe00...	ICMPv6	118	Echo (ping) reply id=0x0001, seq=2, hop limit=64 (request in 5)
7	5.347775	fe80::e19:15fff:febf...	2000::e442:1fff:fe00...	ICMPv6	86	Neighbor Solicitation for 2000::e442:1fff:fe00:4223 from 0c:19:15:bf:00:01
8	5.348457	2000::e442:1fff:fe00...	fe80::e19:15fff:febf...	ICMPv6	78	Neighbor Advertisement 2000::e442:1fff:fe00:4223 (sol)
9	10.427872	fe80::e19:15fff:febf...	fe80::e19:15fff:febf...	ICMPv6	86	Neighbor Solicitation for fe80::e19:15fff:febf:1 from e6:42:01:00:42:23
10	10.431554	fe80::e19:15fff:febf...	fe80::e19:15fff:febf...	ICMPv6	86	Neighbor Advertisement fe80::e19:15fff:febf:1 (rtr, sol)
11	15.506675	fe80::e19:15fff:febf...	fe80::e19:15fff:febf...	ICMPv6	86	Neighbor Solicitation for fe80::e442:1fff:fe00:4223 from 0c:19:15:bf:00:01
12	15.507366	fe80::e442:1fff:fe00...	fe80::e19:15fff:febf...	ICMPv6	78	Neighbor Advertisement fe80::e442:1fff:fe00:4223 (sol)
13	59.817348	fe80::e442:1fff:fe00...	ff02::1::2	DHCPv6	98	Information-request XID: 0x7b23c6 CID: 00030001e64201004223
14	59.875353	fe80::e19:15fff:febf...	fe80::e442:1fff:fe00...	DHCPv6	142	Reply XID: 0x7b23c6 CID: 00030001e64201004223
15	59.833993	2000::e442:1fff:fe00...	2000::1	ICMPv6	118	Echo (ping) request id=0x0001, seq=1, hop limit=64 (reply in 16)
16	59.836393	2000::1	2000::e442:1fff:fe00...	ICMPv6	118	Echo (ping) reply id=0x0001, seq=1, hop limit=64 (request in 15)
17	81.837327	2000::e442:1fff:fe00...	2000::1	ICMPv6	118	Echo (ping) request id=0x0001, seq=2, hop limit=64 (reply in 18)
18	81.839119	2000::1	2000::e442:1fff:fe00...	ICMPv6	118	Echo (ping) reply id=0x0001, seq=2, hop limit=64 (request in 17)
19	85.950258	fe80::e442:1fff:fe00...	2000::1	ICMPv6	86	Neighbor Solicitation for 2000::1 from e6:42:01:00:42:23
20	85.952735	2000::1	fe80::e442:1fff:fe00...	ICMPv6	78	Neighbor Advertisement 2000::1 (rtr, sol)
21	86.244474	fe80::e19:15fff:febf...	2000::e442:1fff:fe00...	ICMPv6	86	Neighbor Solicitation for 2000::e442:1fff:fe00:4223 from 0c:19:15:bf:00:01
22	86.244761	2000::e442:1fff:fe00...	fe80::e19:15fff:febf...	ICMPv6	78	Neighbor Advertisement 2000::e442:1fff:fe00:4223 (sol)
23	91.323834	fe80::e442:1fff:fe00...	fe80::e19:15fff:febf...	ICMPv6	86	Neighbor Solicitation for fe80::e19:15fff:febf:1 from e6:42:01:00:42:23
24	91.329043	fe80::e19:15fff:febf...	fe80::e442:1fff:fe00...	ICMPv6	78	Neighbor Advertisement fe80::e19:15fff:febf:1 (rtr, sol)
25	91.305214	fe80::e19:15fff:febf...	fe80::e442:1fff:fe00...	ICMPv6	86	Neighbor Solicitation for fe80::e442:1fff:fe00:4223 from 0c:19:15:bf:00:01
26	91.366328	fe80::e442:1fff:fe00...	fe80::e19:15fff:febf...	ICMPv6	78	Neighbor Advertisement fe80::e442:1fff:fe00:4223 (sol)
27	101.840060	fe80::e19:15fff:febf...	ff02::1	ICMPv6	118	Router Advertisement from 0c:19:15:bf:00:01

Рисунок 39: Трафик

5.44 Настройка DHCP в случае IPv6

На маршрутизаторе настроим DHCPv6 с отслеживанием состояния. На интерфейсе eth2 маршрутизатора настроим объявления о маршрутизаторах. Опция managed-flag означает, что хосты используют администрируемый (отслеживающий состояние) протокол для автоматической настройки адресов в дополнение к любым адресам, автоматически настраиваемым с помощью SLAAC.

5.45 Настройка DHCP в случае IPv6

```
antoychubekova@antoychubekova-gw-01# set service router-advert interface eth2 ma
naged-flag
[edit]
antoychubekova@antoychubekova-gw-01#
```

Рисунок 40: Настройка объявления о маршрутизаторах

5.46 Настройка DHCP в случае IPv6

Добавим конфигурацию DHCP-сервера на маршрутизаторе. Создаем разделяемая сеть (shared-network-name) с названием username, подсеть (subnet) с адресом 2001::/64, задаем диапазон адресов (range) с именем hosts, содержащий адреса 2001::100 – 2001::199.

На маршрутизаторе посмотрим статистику DHCP-сервера и выданные адреса.

```
antoychubekova@antoychubekova-gw-01# run show dhcpv6 server leases
IPv6 address  State  Last communication  Lease expiration  Remaining
Type      Pool  IAID_DUID
-----
[edit]
antoychubekova@antoychubekova-gw-01#
```

Рисунок 41: Статистика DHCP-сервера

5.47 Настройка DHCP в случае IPv6

Подключимся к узлу PC3 и проверим настройки сети.

```
PC3-antoychubekova console is now available... Press RETURN to get started.
(root@ PC3-antoychubekova)~#
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::28b5:2fff:fe9a:7c29 prefixlen 64 scopeid 0x20<link>
    ether 2a:b5:2f:9a:7c:29 txqueuelen 1000 (Ethernet)
    RX packets 13 bytes 1470 (1.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14 bytes 1076 (1.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::ccbf:9bff:fe9b:2b9d prefixlen 64 scopeid 0x20<link>
    ether ce:bf:9b:9b:2b:9d txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(root@ PC3-antoychubekova)~#
```

Рисунок 42: Настройка сети

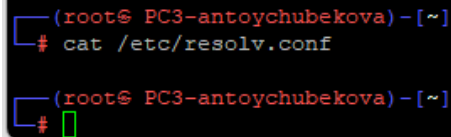
5.48 Настройка DHCP в случае IPv6

```
(root@ PC3-antoychubekova)~#  
# route -n -A inet6  
Kernel IPv6 routing table  
Destination          Next Hop             Flag Met Ref  Use If  
fe80::/64             ::                   U    256 1    0 eth0  
fe80::/64             ::                   U    256 1    0 eth1  
::/0                  fe80::e19:15ff:febf:2 UGD Ae 1024 1    0 et  
h0  
::1/128               ::                   Un   0   2    0 lo  
fe80::28b5:2fff:fe9a:7c29/128 ::                   Un   0   2    0 eth0  
fe80::ccbf:9bff:fe9b:2b9d/128 ::                   Un   0   3    0 eth1  
ff00::/8              ::                   U    256 6    0 eth0  
ff00::/8              ::                   U    256 1    0 eth1  
::/0                  ::                   !n   -1  1    0 lo
```

Рисунок 43: Настройка сети

5.49 Настройка DHCP в случае IPv6

На узле PC3 проверим настройки DNS.



```
(root@ PC3-antoychubekova) - [~]  
# cat /etc/resolv.conf  
  
(root@ PC3-antoychubekova) - [~]  
#
```

Рисунок 44: Настройка DNS

5.50 Настройка DHCP в случае IPv6

На узле PC3 получите адрес по DHCPv.

```
(root@PC3-antoychubekova)-[~]  
# dhclient -6 -v eth0  
Internet Systems Consortium DHCP Client 4.4.3-P1  
Copyright 2004-2022 Internet Systems Consortium.  
All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
  
Listening on Socket/eth0  
Sending on Socket/eth0  
Created duid "\000\001\000\0010\306\224\205\265\232)".  
PRC: Soliciting for leases (INIT).  
XMT: Forming Solicit, 0 ms elapsed.  
XMT: X-- IA_NA 2f:9a:7c:29  
XMT: | X-- Request renew in +3600  
XMT: | X-- Request rebind in +5400  
XMT: Solicit on eth0, interval 1090ms.  
RCV: Advertise message on eth0 from fe80::e19:15ff:febf:2.  
RCV: X-- IA_NA 2f:9a:7c:29  
RCV: | X-- starts 1765005317  
RCV: | X-- t1 - renew +0  
RCV: | X-- t2 - rebind +0  
RCV: | X-- [Options]  
RCV: | | X-- IAADDR 2001::198  
RCV: | | | X-- Preferred lifetime 27000.  
RCV: | | | X-- Max lifetime 43200.  
RCV: X-- Server ID: 00:01:00:01:30:c6:8b:74:0c:19:15:bf:00:01  
RCV: Advertisement recorded.  
PRC: Selecting best advertised lease.  
PRC: Considering best lease.  
PRC: X-- Initial candidate 00:01:00:01:30:c6:8b:74:0c:19:15:bf:00:01 (s: 10105,  
p: 0).  
XMT: Forming Request, 0 ms elapsed.  
XMT: X-- IA_NA 2f:9a:7c:29  
XMT: | X-- Requested renew +3600  
XMT: | X-- Requested rebind +5400  
XMT: | | X-- IAADDR 2001::198  
XMT: | | | X-- Preferred lifetime +7200  
XMT: | | | X-- Max lifetime +7500  
XMT: V IA_NA appended.  
XMT: Request on eth0, interval 940ms.  
RCV: Reply message on eth0 from fe80::e19:15ff:febf:2.  
RCV: X-- IA_NA 2f:9a:7c:29  
RCV: | X-- starts 1765005318  
RCV: | X-- t1 - renew +0
```

5.51 Настройка DHCP в случае IPv6

Вновь на узле PC3 проверим настройки сети, пропингуем маршрутизатор, проверьте настройки DNS.

```
(root@ PC3-antoychubekova)~#  
# ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 2001::198 prefixlen 128 scopeid 0x0<global>  
    inet6 fe80::28b5:2fff:fe9a:7c29 prefixlen 64 scopeid 0x20<link>  
    ether 2a:b5:2f:9a:7c:29 txqueuelen 1000 (Ethernet)  
    RX packets 17 bytes 2014 (1.9 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 21 bytes 1828 (1.7 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::ccbf:9bff:fe9b:2b9d prefixlen 64 scopeid 0x20<link>  
    ether ce:bf:9b:9b:2b:9d txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(root@ PC3-antoychubekova)~#
```

5.52 Настройка DHCP в случае IPv6

```
(root@ PC3-antoychubekova)~#  
# route -n -A inet6  
Kernel IPv6 routing table  
Destination                Next Hop                    Flag Met Ref  Use If  
2001::198/128                ::                          Ue    256 1      0 eth0  
fe80::/64                    ::                          U      256 1      0 eth0  
fe80::/64                    ::                          U      256 1      0 eth1  
::/0                         fe80::e19:15ff:febf:2       UGDae 1024 1      0 et  
h0  
::1/128                      ::                          Un     0 3      0 lo  
2001::198/128                ::                          Un     0 2      0 eth0  
fe80::28b5:2fff:fe9a:7c29/128 ::                          Un     0 5      0 eth0  
fe80::ccbf:9bff:fe9b:2b9d/128 ::                          Un     0 3      0 eth1  
ff00::/8                     ::                          U      256 6      0 eth0  
ff00::/8                     ::                          U      256 1      0 eth1  
::/0                         ::                          !n    -1 1      0 lo  
  
(root@ PC3-antoychubekova)~#
```

Рисунок 47: Настройка сети

5.53 Настройка DHCP в случае IPv6

```
(root@ PC3-antoychubekova)~[~]  
# ping 2001::1 -c 2  
PING 2001::1 (2001::1) 56 data bytes  
64 bytes from 2001::1: icmp_seq=1 ttl=64 time=16.6 ms  
64 bytes from 2001::1: icmp_seq=2 ttl=64 time=7.12 ms  
  
--- 2001::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 100lms  
rtt min/avg/max/mdev = 7.120/11.847/16.574/4.727 ms  
  
(root@ PC3-antoychubekova)~[~]  
#
```

Рисунок 48: Проверка достпности

5.54 Настройка DHCP в случае IPv6

```
(root@ PC3-antoychubekova) -[~]  
# cat /etc/resolv.conf  
search antoychubekova.net.  
nameserver 2001::1  
  
(root@ PC3-antoychubekova) -[~]  
#
```

Рисунок 49: Настройка DNS

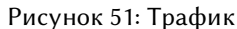
5.55 Настройка DHCP в случае IPv6

На маршрутизаторе посмотрив статистику DHCP-сервера и выданные адреса.

```
antoychubekova@antoychubekova-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication   Lease expiration      Remaining
Type              Pool                               IAID_DUID
-----
2001::198         active    2025/12/06 07:15:18   2025/12/06 09:20:18   2:02:16
non-temporary    antoychubekova-stateful  29:7c:9a:2f:00:01:00:01:30:c6:94:85:2a:
b5:2f:9a:7c:29
[edit]
antoychubekova@antoychubekova-gw-01#
```

Рисунок 50: Статистика DHCP-сервера

Далее посмотри захваченный трафик по DHCPv6.



Раздел 6

6. Выводы

6.1 Выводы

В ходе выполнения лабораторной работы №7 я получила навыки настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.